Surveys Based on Time Frame

Type	Description	Example
Cross-sectional Surveys	One-time data collection	Social media usage trends
Longitudinal Surveys	Repeated data collection over time	COVID-19 behavior tracking
Panel Surveys	Following the same individuals over time	Customer loyalty surveys

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What Is a Cross-Sectional Survey?

- A **cross-sectional survey** is a research method used to collect data from a **specific population at a single point in time**.
- It provides a **snapshot** of opinions, behaviors, or characteristics but does not track changes over time.

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Features of Cross-Sectional Surveys

- One-Time Data Collection
- Descriptive in Nature
- Quick
- Used in Various Fields

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Advantages of Cross-Sectional Surveys

- Efficient and Time-Saving
- Large-Scale Data Collection
- No Risk of Dropout
- Useful for measuring how common a behavior

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Limitations of Cross-Sectional Surveys

- **Cannot Track Changes Over Time**
- **Possible Response Bias**
- **No Cause-and-Effect Relationship**

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Cass-Study

- A researcher wants to understand the relationship between **social media use and concern levels** in teenagers.

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Cass-Study

- A researcher wants to understand the relationship between **social media use and concern levels** in teenagers.
- **Method:**
  - A survey is conducted among **10,000 high school students** in different cities, asking:
    - "How many hours do you spend on social media daily?"
    - "On a scale of 1-10, how anxious do you feel daily?"

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Designing a Cross-Sectional Survey

- **Research Topic:**
- **"The Impact of Screen Time on Student Productivity"**
  - **Step 1: Define the Research Objective**
  - **Step 2: Identify the Target Population**
  - **Step 3: Choose the Data Collection Method**
  - **Step 4: Design the Survey Questions**
  - **Step 5: Conduct the Survey in Class**
  - **Step 6: Analyze & Discuss the Results**

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What Is a Longitudinal Survey?

- A **longitudinal survey** is a research method used to collect data **from the same individuals or groups over an extended period**.
- Unlike cross-sectional surveys, which provide a **snapshot**, longitudinal surveys track **changes and trends over time**.

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Features of Longitudinal Surveys

- **Tracks Changes over Time**
- **Repeated Data Collection**
- **Identifies Cause-and-Effect Relationships**
- **Used in Multiple Fields**

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Types of Longitudinal Surveys

- **Cohort Study**
  - Tracking **the career paths of graduates from the class of 2024**.
- **Trend Study**
  - A survey measuring **public trust in the government** every five years.

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Advantages of Longitudinal Surveys

- **Better for Studying Changes & Trends**
- **Establishes Cause-and-Effect**
- **More Accurate Data**

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Limitations of Longitudinal Surveys

- Time-Consuming and Expensive
- Participant Dropout
- Data Collection Challenges

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Case Study

- Example: Longitudinal Survey on Student Performance

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Case Study

- **Example: Longitudinal Survey on Student Performance**
  - **Objective:**
    - Track **how study habits affect GPA** over four years of university.
  - **Method:**
    - **Freshman Year:** Ask students about their **study habits** (hours studied, preferred learning methods).
    - **Sophomore Year:** Follow up with the same students to check **GPA changes**.
    - **Junior Year:** Assess if students **improved or changed their study habits**.
    - **Senior Year:** Compare final GPA with **initial study habits** to identify trends.

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Designing a Longitudinal Survey

- **Research Topic:**
- **"How Study Habits Affect Academic Performance Over a Semester"**

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Designing a Longitudinal Survey

- **Research Topic:**
- **"How Study Habits Affect Academic Performance Over a Semester"**
  - **Step 1: Define the Research Objective**
  - **Step 2: Identify the Target Population**
  - **Step 3: Choose the Data Collection Method**
  - **Step 4: Design the Survey Questions**
  - **Step 5: Conduct the Surveys at 3 Key Points**
  - **Step 6: Analyze & Discuss the Results**

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What is a Panel Survey?

- A **panel survey** is a type of **longitudinal survey** where the **same group of individuals is surveyed multiple times over a period** to observe changes in their opinions, behaviors, or conditions.

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Features of Panel Surveys

- Same Participants Over Time
- Multiple Survey Waves
- Tracks Individual-Level Changes

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Types of Panel Surveys

- Continuous Panel Survey
- Discontinuous Panel Survey

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Advantages of Panel Surveys

- Tracks Behavioral Changes Over Time
- More Accurate & Reliable
- Helps predict future trends

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Limitation of Panel Surveys

- Time-Consuming
- Panel Dropout
- Bias Risk

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Designing Panel Survey

- **Research Topic:**
  - Track **stress levels and coping strategies** among university students throughout the semester.

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